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Transtibial Prosthetic Socket Strength: The Use of ISO 10328 in the Comparison of Standard and 3D-Printed Sockets

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ABSTRACT

Introduction: Prosthetic sockets must withstand instances of maximum loading without failure. The goal of the present study is to evaluate the strength at failure and the failure mechanism for 3D-printed transtibial sockets, thermoplastic transtibial check sockets, and carbon-fiber definitive laminated transtibial sockets.

Materials and Methods: Three clinically available materials (carbon fiber, PETG thermoplastic, and 3D print polylactic acid [PLA]) were used to produce identically shaped sockets. In final assembly, the carbon-fiber sockets were designed for use both with and without a pin-lock system. Thermoplastic sockets and 3D-printed sockets were designed for use without a pin-lock system. These socket systems were then tested following International Standards Organization (ISO) 10328. Ultimate strength (US) at failure, maximum deflection at failure, and failure mechanism were determined.

Results: Both designs of carbon-fiber sockets had higher US values at failure than thermoplastic sockets and 3D-printed PLA sockets. Thermoplastic sockets had a slightly higher average US than 3D-printed sockets, but 3D-printed sockets exhibited a greater strength-to-weight ratio. Four distinct socket failure mechanisms were determined.

Conclusions: Failure mechanisms and US values differed for all socket types. Carbon-fiber sockets exhibited the highest US, but 3D-printed PLA sockets showed comparable strength to current thermoplastic sockets. Although the ISO 10328 testing standard was sufficient to complete these evaluations, the method lacked some socket-specific measures and loading conditions that could have improved comparisons between socket types. (*J Prosthet Orthot.* 2020;32:93–100)

KEY INDEXING TERMS: transtibial prosthetic sockets, 3D printing, fabrication methods, International Standards Organization 10328, ultimate strength, static proof

A prosthetic socket must be designed and fabricated to ensure appropriate fit, comfort, strength, and durability. With the use of clinical studies and patient surveys, the evaluation of fit and comfort is often reported in the literature,[1–3] but much less research has been published on the quantification of prosthetic socket strength or durability. Structural testing of lower-limb prosthetic systems can provide insight on the behavior of the system under load and the overall strength of the system, but few studies have evaluated the strength of these systems before in-patient use.

In 1987, Wevers and Durance[4] developed a dynamic testing machine designed to replicate loads experienced during a normal walking gait and used the machine to both statically and dynamically test transtibial prosthetic sockets. The testing machine was designed based on preliminary standards set forth by the International Society for Prosthetics and Orthotics (ISPO). Later, Coombes and MacCoughlan[5] used similar ISPO standards criteria to conduct static and cyclic testing of different prosthetic shanks at a load of 2500 N. After the introduction of the ISO 10328 standard (prosthetics–structural testing of lower-limb prostheses–requirements and test methods) in 1996, structural testing of lower-limb prosthetic sockets has been conducted using these guidelines.[6] Although not explicitly written for socket strength testing, several groups have replicated the testing conditions to conduct strength testing of lower-limb prosthetic sockets. In 1999, Current et al.[7] examined the static structural integrity of different socket types with varying reinforcement materials and resins. A socket loading fixture was developed to apply loading to the socket following the ISO 10328 standard. All 10 sockets failed to pass the standards for loading at level A100 (ultimate static test force of 4025 N, P5 level in revised standard) for loading condition II, but the authors were able to establish a consistent testing method. More recently, Gerschutz et al.[8] conducted socket strength testing following the ISO standard. Socket material type, fabrication method, and fabrication facility were evaluated, and each was found to influence socket strength. Ultimate socket failure loads ranged from 474 N to 5,713 N depending on these factors.

Recently, many groups have explored additive manufacturing methods, specifically 3D printing, for relevance in producing a lower-limb prosthetic socket. Traditional prosthetic socket

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fabrication methods are time-consuming, requiring several independent steps that must be repeated if the socket fit is incorrect. Many of the steps associated with fitting and fabrication require materials that are ultimately discarded. Even with the use of CAD/CAM (computer-aided design/computer-aided manufacturing) methods to replace traditional plaster casting, the positive limb mold required for socket shape is either discarded or must be stored requiring space in clinic or at a central fabrication (CFAB) facility. In addition, there are noted inconsistencies in shape between sockets fabricated traditionally.[8] The 3D printing technology poses great potential to eliminate many issues associated with traditional socket fabrication processes.

For the last 20 years, increased focus has been placed on the design and development of 3D-printed sockets, using such methods as selective laser sintering (SLS) and fused deposition modeling (FDM).[9] These groups have had notable success; however, little work has been conducted to quantify the strength of prosthetic sockets produced using additive manufacturing methods. Specifically, there is limited prior work examining the strength of these sockets using ISO 10328 methods, although there have been several studies in which additive manufactured socket durability was evaluated in-patient.[10,11] Rogers et al.[10] used solid freeform fabrication to print a compliant prosthetic socket and evaluated durability during a year of use by a single patient. The study did not address long-term durability of the additively manufactured socket. The Squirt Shape socket, developed at Northwestern University, is produced using extrusion-deposition techniques.[11] The materials were analyzed for strength and fatigue, but full sockets were not tested. The Squirt Shape sockets were evaluated using long-term clinical fittings, and a 20-month period without failure was reported at the time of publication.[11]

Using an in-house developed method known as the rapid socket manufacturing machine (RSMM), Goh et al.[12] used ISO 10328 standards to study the structural integrity of sockets fabricated with this method. RSMM sockets were tested following the static proof test guidelines and the ultimate strength (US) test guidelines for load condition II. Permanent deformation after static testing was within the acceptable range, and the socket was able to withstand the ultimate test load of 4.025 kN without failure and with 12 mm total of permanent deformation. The additively manufactured RSMM socket successfully completed cyclic loading to 250,000 cycles. Aside from this study, there have been no other published results of additive manufactured socket strength when tested following ISO 10328 standards.

The purpose of the present study was to evaluate further the strength of additively manufactured and traditionally fabricated lower-limb prosthetic sockets when tested following ISO 10328. This initial study determined comparative US of different socket types and evaluated the effectiveness of the ISO 10328 standard for testing both traditional and 3D-printed socket strength.

METHODS

Three socket types were chosen for this study: carbon-fiber definitive laminated sockets (CFs), thermoplastic check sockets (TPs), and 3D-printed sockets (3Ds).

SOCKET FABRICATION

All sockets were fabricated based on a single-patient deidentified limb model from a 5′6″, 186-pound male subject. The limb model had a cylindrical/conical shape with no abnormal protrusions or features requiring extensive design modifications. A 3D scan of the residual limb was modified by a certified prosthetist to produce a transtibial socket shape, and the socket model was provided to three fabrication facilities. Five carbon-fiber definitive laminated sockets were fabricated by each of two different fabrication facilities (facilities A and B). Each facility was provided with the same information and limb model, and was asked to supply both the socket and the distal plate, pyramid, and pylon for each socket for the study. The fabrication method was determined by the individual facility. Facility A used a carbon-fiber braider to produce the definitive laminated sockets. Socket fabrication for facility B consisted of laminating the limb model with carbon fiber and resin and tying the carbon fiber into the distal plate. Five 3D-printed sockets were printed by a single fabrication facility (facility C). Again, the facility was provided with the same information and limb model and determined the print methods and materials. All sockets were printed in polylactic acid (PLA) using an FDM printing method (Rostock Max v3 Printer). Socket modeling (Omega and Blender) and slicing (Simplify3D) were completed by the fabrication facility. Overall wall thickness was set to 7 mm with a 40% infill and 0.4-mm print layer height. Attachment components for the 3D-printed sockets were provided by the facility. A single set of five thermoplastic PETG check sockets was provided by one facility (facility B), fabricated using half-inch (12.7-mm) thick PETG thermoplastic sheets pulled over the limb model. Average socket wall thickness at specific locations was determined after fabrication. Average lateral flare thickness, medial flare thickness, and posterior wall thickness were measured before testing and averaged 5.41 mm (±0.35 mm), 5.44 mm (±0.34 mm), and 4.04 mm (±0.84 mm), respectively. Attachment components for the thermoplastic sockets were provided by the fabrication facility. Table 1 lists the sockets types along with the attachment components for each socket type.

Sockets fabricated by facility A were designed for a pin-lock suspension allowing for the placement of a Bulldog Lock at the distal end. The pin-lock design was hypothesized to be a worst-case scenario. Failure of prosthetic sockets has been previously observed at the distal end.[4,7,8] With a pin-lock design, the distal end is extended and sandwiched between the lock and the pyramid. Sockets fabricated by facilities B and C were designed for use without a pin-lock system. The distal plates for the thermoplastic check socket and the 3D-printed socket were attached to the distal end using epoxy. Each socket was then fiberglass wrapped to a standard socket wall height by a single prosthetist. Failure was hypothesized to occur at the epoxy for the PETG check sockets and the 3D-printed sockets.

To apply a load into the socket, a 30-mm solid aluminum mandrel was cast in body filler to create a limb dummy. The limb dummy was cast so that a 5-cm distal void would be present in the socket. The mandrel was aligned following methods modified from those developed by Neo et al.[13] The socket was attached to the bottom load application lever and placed in the alignment jig for alignment of the mandrel. Once aligned, body filler was

Table 1. Summary of socket samples and attachment components provided by the respective facilities

Socket Type	Facility Type	Abbreviations	Attachment Components
Carbon-fiber definitive laminated socket	CFAB (A)	CFA	Bulldog standard push lock 4-hole male pyramid titanium 30-mm aluminum pylon
Carbon-fiber definitive laminated socket	CFAB (B)	CFB	Ossur L-54440 plate 4-hole male pyramid titanium 30-mm female pylon (carbon fiber/titanium)
Thermoplastic check socket	CFAB (B)	TPC	Ossur L-54440 plate 4-hole male pyramid titanium 30-mm female pylon (carbon fiber/titanium)
3D-printed socket	3D print facility (C)	3DC	Ossur L-54440 plate 4-hole male pyramid titanium 30-mm female pylon (carbon fiber/titanium)

CFAB indicates central fabrication facility; CFA, carbon-fiber definitive laminated from facility A; CFB, carbon-fiber definitive laminated from facility B; TPB, thermoplastic check socket from facility B; and 3DC, 3D-printed socket from facility C.

poured and allowed 24 hrs to harden. The same limb dummy was used for all tests to maintain consistency. The limb dummy was inspected after each test for signs of damage, and any fit deviations were noted with each test.

SOCKET TESTING

ISO 10328 standards were used to guide static testing of the sockets. The standard specifies loading conditions, both magnitude and location, that are representative of specific instances of a normal gait cycle, specifically early stance phase (heel strike) and late stance phase (toe-off). An Instron 8874 bi-axial servohydraulic testing system (Illinois Tool Works Inc., Norwood, MA, USA) with a 25-kN load cell was used to apply loads to the socket samples. Figure 1 shows the complete test setup for a late toe-off test position. To apply the specified loads at the correct locations, a special attachment fixture was developed. The attachment fixture consisted of a top loading fixture and a base loading fixture (Figure 2). The fixtures were designed based on the socket loading fixture presented by Current et al. and the loading plates presented by Neo et al. The base fixture has pretapped holes that allow for attachment of a 2-inch diameter ball joint. The correct base hole location is determined by the loading condition and socket type (left or right). The top fixture attaches to the mandrel embedded in the limb dummy and allows for adjustment of the top loading point, which consists of a one-and-one-half-inch diameter ball joint. The slot is marked at the appropriate lengths for loading condition I (early heel strike) and loading condition II (late toe-off) to allow for accurate and rapid alignment.

All sockets underwent static proof testing and US testing. The ISO standard defines proof strength as a single instance of severe load, which the prosthetic device can sustain and still function as intended after. US is defined as a single instance of severe load, which the prosthetic device can sustain but the load might render the device unusable, either due to permanent deformation greater than that specified by proof loads or total device failure. Socket samples were first tested to level P7 static proof loads for load condition I and level P5 static proof loads for load condition II. Sockets were loaded at a constant rate to a set force, 2900 N for load condition I and 2013 N for load condition II; the applied force was held for 30 seconds and then the socket was unloaded at a constant rate. After static proof testing, sockets were tested to failure. Socket failure was deemed the point at which the socket/socket system could no longer support increasing loads. Figure 3 diagrams the order for testing. All sockets were

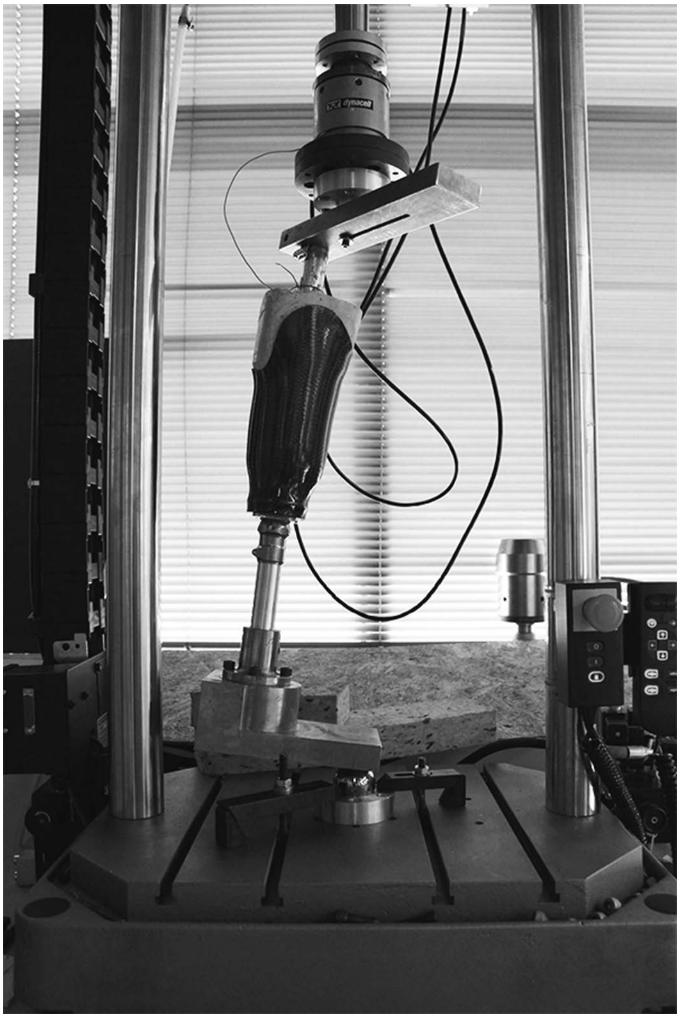

Figure 1. Testing configuration for static structural testing following ISO 10328 for load configuration II.

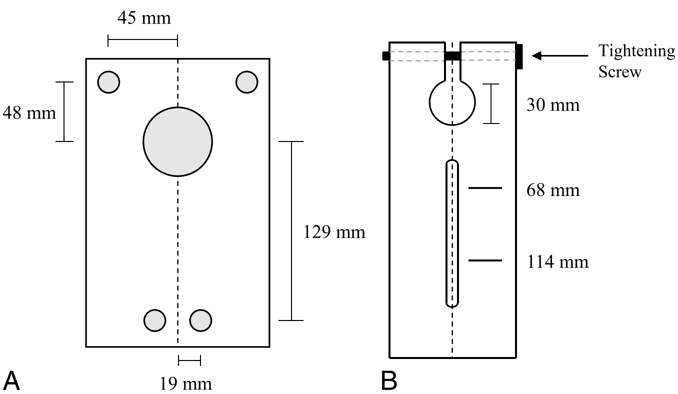

Figure 2. Schematics for load application levers. A, Bottom load application lever with four attachment locations for ball joint. The distal hole locations are at correct distances for load condition I testing. The proximal hole locations are correct distances for load condition II testing. B, Top load application lever attaches to the mandrel of the limb dummy via a 30-mm clamp gap. A slot allows for quick adjustment of the top ball joint.

examined between tests for signs of visual damage. Sockets were tested in all conditions independent of the results of the previous test condition. For example, if a socket failed testing under the ISO standards for load condition I, the socket was still tested in load condition II and then to failure. All tests were conducted at a loading rate of 100 N/s as specified by the ISO standard. Deviating from the standard, all sockets were aligned to a neutral position rather than a worst-case alignment scenario to eliminate additional confounding variables during analysis. Total segment length (distance between bottom load application point and top load application point) was recorded before and after static proof testing for both loading conditions. During US testing, failure was identified as a decrease of 40% or greater in load supported in consecutive measurements or total socket failure due to break.

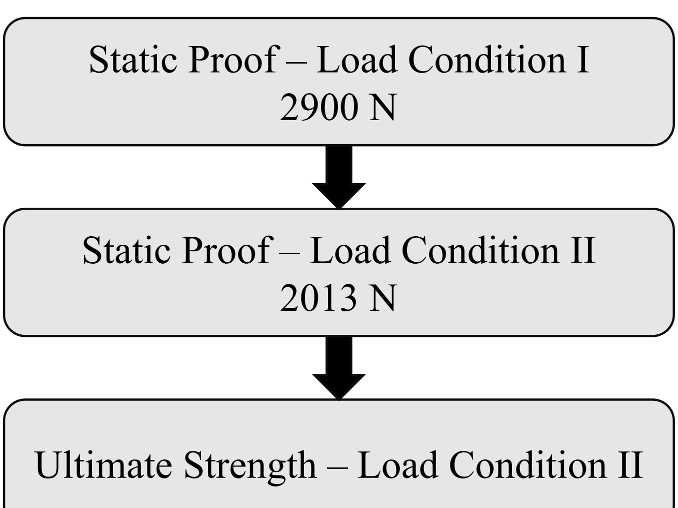

Figure 3. Testing followed standard protocol set forth by ISO 10328. Each socket was tested in load condition I, followed by load condition II, and finally to failure (ultimate strength) in load condition II. Each socket was allowed a minimum 24-hr rest period between testing scenarios.

RESULTS

SOCKET WEIGHTS

Before testing, sockets were weighed with distal plates or Bulldog Locks. Table 2 reports the socket weights. CFA sockets had an average weight of 287.62 g (±6.86 g). CFB sockets had an average weight of 474.68 g (±7.96 g). 3DC sockets had an average weight of 661.01 g (±18.98 g). TPB sockets had an average weight of 557.83 g (±28.50 g).

STATIC PROOF TESTING

All CFA and CFB sockets passed static proof testing for both loading conditions. CFA sockets had an average total deformation of 0.60 mm after P7 load condition I testing and 1.10 mm after load condition II testing. CFB sockets had no recorded deformation after P7 load condition I testing and 0.02 mm after P5 load condition II testing. All TPB sockets passed static proof testing for both loading conditions I and II with average total deformations 0.08 cm and 0.20 cm, respectively. 3DC sockets passed static proof testing for both P7 load condition I and P5 load condition II testing. All other samples showed no visible signs of damage after static proof testing.

Socket deflection was defined as the distance of crosshead movement of the system indicating deflection of the socket test system (socket, pylon, attachment fixtures). The system reached maximum deflection during the hold period of static proof loading for both load conditions. CFA sockets experienced an average maximum deflection of 3.4 mm (±0.2 mm) in load condition I and 8.0 mm (±1.5 mm) in load condition II. CFB sockets had an average maximum deflection of 3.9 mm (±0.25 mm) in load condition I and 5.4 mm (±0.4 mm) in load condition II. TPB sockets had an average maximum deflection of 5.7 mm (±0.3 mm) for load condition I and 10.4 mm (±0.5 mm) for load condition II. Average maximum deflections for 3DC sockets were 3.4 mm (±0.3 mm) and 7.0 mm (±0.4 mm) for load conditions I and II, respectively.

Table 2. Each socket sample was weighed before testing

Sample	Average Socket Weight, g	Average Strength-to-Weight Ratio, N/g
CFA	287.62 (6.86)	21.53 (2.97)*
CFB	474.13 (7.96)	13.63 (0.22)
TPB	557.83 (28.50)	7.34 (0.75)
3DC	661.01 (18.98)	5.81 (0.74)

TPB and 3DC socket weights include distal plate attached with epoxy and fiberglass. CFA and CFB sockets weights include the Bulldog Lock or distal plate, respectively. After ultimate strength testing, a strength-to-weight ratio for each socket was calculated by dividing the force at failure by the socket weight. Reported as average (standard deviation).

*The strength-to-weight ratio for CFA includes only those sockets that experienced total failure.

3D indicates 3D-printed socket; CF, carbon-fiber definitive laminated socket; TP, thermoplastic check socket; A, facility A; B, facility B; C, facility C.

ULTIMATE STRENGTH TESTING

The compressive load at failure was deemed the US of the socket and was defined as the point at which the socket system could no longer support increasing load. Table 3 lists the overall results matrix for all sockets tested. The US value for each socket was recorded as a single maximum load at failure. Failure of the CFA sockets was considered total system failure. Although the socket experienced catastrophic failure at the US load, the pylon and lock system also failed in conjunction. The average US for CFA sockets was 5575 N (±1039.7 N). Excluding those sockets that experienced component failure before any signs of socket failure, the average US increases to 6727.6 N (±524.25 N). CFB sockets showed no visible signs of damage after US testing. For all CFB sockets, the pylon failed before any socket failure occurred, and average US of the socket system was recorded as 6462 N (±74.79 N). Average maximum deflection values for CFA and CFB sockets at US load were 34.93 (±1.18 mm) and 31.36 mm (±1.64 mm), respectively. Figure 4 depicts graphically the US values for each socket type.

TPB sockets had an average US of 4091.78 N (±477.30 N). Average maximum deflection at US load for TPB sockets was 23.05 mm (±2.92 mm). Average US for TPB sockets was 30.7% lower than that for CFA sockets and 44.9% lower than that for CFB sockets. There were visible signs of failure at the distal end of the interior socket. Because the thermoplastic material is clear, the epoxy can be seen at the distal end, and for each socket sample, there were visible cracks between the epoxy and the socket.

3DC sockets had an average US of 3836.85 N (±478.38 N) and an average maximum deflection of 16.90 mm (±3.01 mm). The average US for 3DC sockets was 36.94% lower than that for CFA sockets and 51.0% lower than that for CFB sockets. Four 3DC socket samples failed at the epoxy-distal plate connection point. For these samples, the socket remained intact, but with an average permanent deformation due to shifting at the distal socket-plate connection. One 3DC socket sample experienced complete socket failure just above the fiberglass wrap with a complete break resulting is two separate socket pieces.

Strength-to-weight ratios were calculated for each socket by dividing the maximum load at failure by the socket weight.[14] Maximum strength-to-weight ratios were found for the definitive laminated sockets, with an average value of 21.53 N/g for CFA and an average of 13.63 N/g for CFB. The 3D-printed sockets had reduced strength-to-weight ratios, with an average of 5.81 N/g for 3DC. TPB sockets had an average strength-to-weight ratio of 7.34 N/g.

DISCUSSION

Three different transtibial prosthetic socket materials and two different socket designs were tested for static strength guided by the ISO 10328 standards. Carbon-fiber definitive laminated sockets, thermoplastic check sockets, and 3D-printed sockets were fabricated for use without a pin-lock system, and a second set of carbon-fiber definitive laminated sockets was fabricated for use with a pin-lock suspension system. The overall goal of the study was to use the ISO 10328 standard to determine the US, or the maximum load at failure, of the socket types and materials as well as the failure mechanisms of the different socket types. Although the long-term durability of a socket is critical, it is important to first understand how a new material or socket design behaves without requiring a patient. Benchtop testing, following ISO standards, provides a method to analyze socket behavior under relevant loading conditions. Standardized benchtop methods show increased relevance with the introduction of new materials and new socket fabrication methods, such as 3D printing. As 3D printing has become a more common practice in orthotics and prosthetics, evaluating the efficacy of these methods for clinical use becomes increasingly important. A common argument against the use of 3D printing for the fabrication of a lower-limb prosthetic socket is the assumed reduced strength of the materials and printing methods; however, there is little published data to test the validity of this statement. Results from the present study add to the published body of knowledge of traditional socket strength and provide new information of the strength and failure mechanisms of 3D-printed sockets.

Table 3. Overall completion matrix for all socket types

Sample	Socket Type	Ultimate Strength, N	Maximum Deflection at Failure, mm	Failure Mode
CFA	Pin-lock style, braided carbon-fiber definitive laminated socket	5575.4 (1039.73)	34.93 (1.18)	Compressive failure/component failure
CFB	Carbon-fiber definitive laminated socket	6462.26 (74.79)	31.36 (1.64)	Component failure (pylon only)
TPB	PETG thermoplastic check socket, distal plate attached with epoxy and fiberglass wrap	4091.78 (477.30)	23.05 (2.92)	Connection failure
3DC	3D-printed socket with distal plate attached with epoxy and fiberglass wrap	3836.85 (478.37)	16.90 (3.01)	Connection failure/tensile failure

The ultimate strength, maximum deflection at failure, and the failure mode are reported for each socket type. Ultimate strength is the maximum load at failure. Maximum deflection is the total cross head extension at failure, representing the overall deflection of the socket at maximum loading. Failure mode is classified as one of four types of failure. Ultimate strength values and maximum deflection values reported as average (standard deviation).

3D indicates 3D-printed socket; CF, carbon fiber definitive laminated socket; TP, thermoplastic check socket; A, facility A; B, facility B; C, facility C.

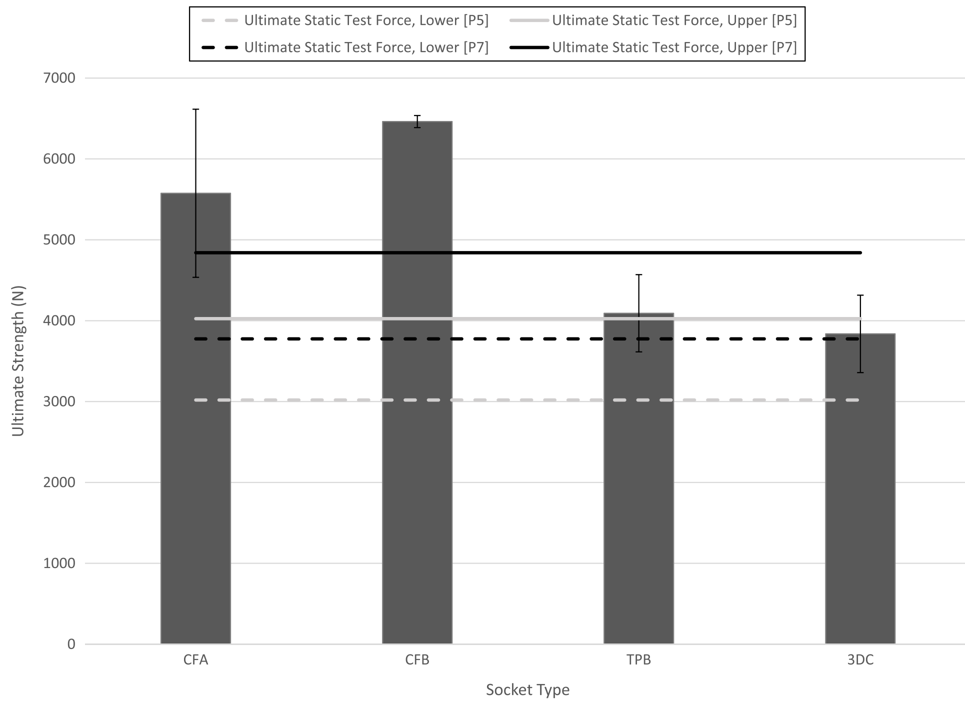

Figure 4. ISO specifies an upper and lower ultimate static test force value for each test level. The figure shows the average ultimate strength of each socket type. The horizontal bars indicate the ISO specified ultimate static test force values. Abbreviations: carbon-fiber definitive laminated from facility A (CFA), carbon-fiber definitive laminated from facility B (CFB), thermoplastic check socket from facility B (TPB), and 3D-printed socket from facility C (3DC).

SOCKET WEIGHT

The weight of a prosthetic socket can be a crucial factor for a prosthetic user. Although there have been limited proven effects of increased prosthetic socket weight on metabolic energy costs or gait,[15,16] patients could have a preference for the weight of the prosthetic socket. Differences in the weight of a prosthetic socket can also be indicative of differences between the sockets. More materials could result in a heavier socket weight. Greater variation in socket weights were noted for those sockets fabricated for use without pin-lock (facilities B and C), especially for the 3D-printed sockets and thermoplastic check sockets. These sockets required the distal plates to be attached to the socket using an epoxy and then wrapped with fiberglass. In addition, traditional manufacturing methods are completed by a technician, and slight variation between the sockets could have been introduced during the labor-intensive traditional fabrication steps.

SOCKET FAILURE BEHAVIOR

The overall goal of the study was to evaluate the behavior of different socket types under standard specified loading. Based on prior published results and knowledge of the loading mechanism, the authors hypothesized that structural failure would occur at the distal end of the socket for each socket type and material. Loads applied for load condition II cause the distal region of the socket to undergo greater bending. Failure at the distal end or below was observed for all sockets, excluding one 3D-printed socket sample. Total crosshead extension at failure was determined for each socket and is indicative of total socket deflection under load. However, during US testing, pylon bending was observed in conjunction with socket failure for carbon-fiber definitive laminated samples, and thus it was difficult to determine the amount of displacement due to socket deformation alone. Similar pylon failure during US testing was reported by Goh et al. Observed failure mechanisms varied between the carbon-fiber definitive laminated sockets, thermoplastic check sockets, and the 3D-printed sockets. CFA sockets failed under a compressive collapse at the anterior distal wall. The maximum loads at failure differ slightly from those reported in literature,[8] but similar failure mechanisms were observed.

During US tests for CFB, TPB, and 3DC sockets, a 30-mm solid aluminum rod was used to simulate the pylon. Even with the change in pylon, CFB sockets did not experience a defined failure. All tests for carbon-fiber definitive laminated sockets were suspended due to pylon failure at approximately 6500 N. At this point, no visible signs of failure in the socket were detected. The maximum load sustained by the socket system was much greater than 5300 N, the upper P7 ultimate static force value given by the ISO standard. The failure mechanism observed indicates that component failure would occur before socket failure for carbon-fiber sockets of this type.

Ultimate socket strength values for the TPB and 3DC sockets were lower than those for the carbon-fiber sockets but were similar, and both were much lower than the ultimate socket strength values for both carbon-fiber socket types. Failure for all thermoplastic sockets occurred below the upper ultimate static test force for load level P7 (4840 N). 3DD sockets failed below the upper ultimate static test force for load level P5 (4025 N). However, in each instance, the socket itself did not fail, but rather the connection between the socket and the distal plate. Although failure was not considered catastrophic because the socket in most cases remained intact, there was distinct failure noted at the connection between the socket and the distal plate. Large values of permanent deformation for thermoplastic and 3D-printed sockets were observed after US testing.

Four distinct failure mechanisms were noted: socket material compressive failure, socket material tensile failure, socket-component connection failure, and component failure (Figure 5). Material failure was noted when a specific part of the socket failed, and this failure could be attributed to the loads (tensile, compressive, or shear) at this specific location. Component or socket-component connection failure was noted when these parts of the system were observed to fail. CFA sockets experienced a material compressive failure mode at the anterior distal wall as well as component failure of the lock and pylon. The base of the socket collapsed into the distal pyramid causing failure. Although failure of the socket, pylon, and Bulldog Lock all occurred during the US test, the distal pyramid adapter did not experience any failure. CFB sockets experienced only component failure due to bending of the pylon. TPB sockets experienced connection failure at the plate/socket connection. 3DC sockets, with the exception of one socket sample, experienced connection failure between the socket and the distal plate similar to the TPC sockets. One 3DC socket experienced complete socket material tensile failure. The socket fractured completely above the fiberglass wrap. The ISO loading places the distal end of the socket in bending leading to a tensile failure at the posterior wall above the fiberglass wrap for the specific 3D-printed socket sample.

Failure of the prosthetic socket can be influenced by a variety of design features related to the socket system. Both socket design and material type contribute to the final structural strength of a socket. As an example, increases in wall thickness will likely contribute to improvements in socket strength at locations where material failure is being observed. However, this increase in material thickness comes at the expense of weight, fabrication complexity, and material cost. For 3D-printed sockets, a variety of design inputs can be quickly changed, most notably print thickness and internal wall geometry. By adjusting different print material and structural properties and making use of testing standards such as the ISO 10328, 3D printing fabrication could offer significant versatility in desired socket performance.

There are several important limitations in the present study that must be noted. Primarily, the ISO standard is not explicitly written for testing prosthetic sockets. The standard was developed to standardize the testing of other components required for use with a prosthetic socket, including pylons, pyramid adapters, ankle-foot devices, and so on. The ISO standard is, however, a reasonable approximation for initial socket testing, in that the load magnitudes and application lines given simulate relevant gait cycle conditions, early heel strike, and late toe-off. Furthermore, following these standards allows for comparison to prior work.[4,7,8,12,13] However, because the test is not designed specifically for socket testing, interpretation of the standard allows for variations in test fixture methods. In previous work, the ISO standard was interpolated to provide offsets for the load application point at the base of the socket ridding the need for a pylon, such as the testing method developed by Gerschutz et al.[8] Without a pylon, loads can be transferred directly to the socket,

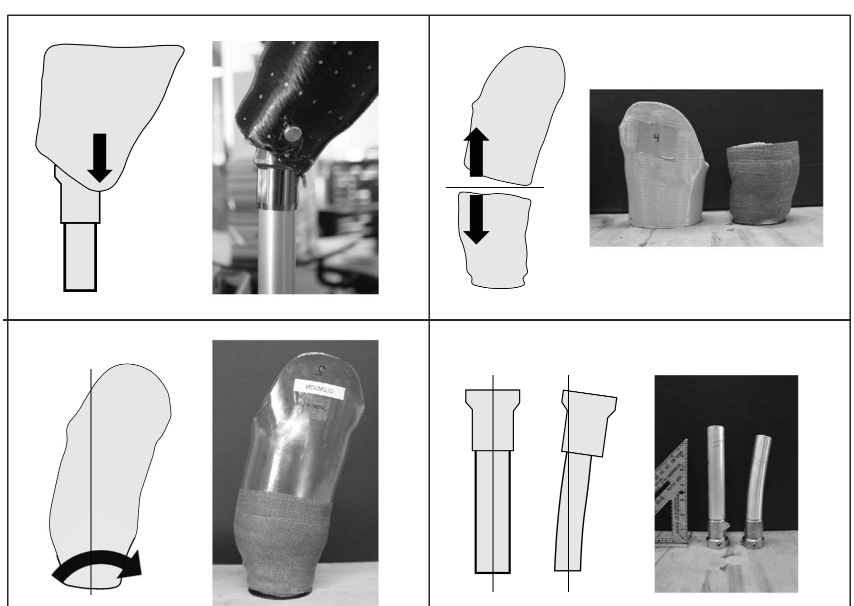

Figure 5. Diagrams depicting socket failure mechanisms along with images of socket samples. A, Compressive failure: the front wall of the socket collapses into the distal pyramid. B, Complete tensile failure: complete break through the entire socket. This failure mechanism only occurred for one 3D-printed socket sample (C). Connection failure: failure at the connection between the distal plate and socket, specifically at the epoxy connection for thermoplastic and 3D-printed sockets. D, Component failure: failure of a component associated with the socket system, such as pylon bending.

and true US of the isolated socket can be determined. For test designs that simulate the full prosthetic system, such as the present study, the effects of loading on the pylon limit the transfer of load directly to the socket. By using the ISO standard, the user can simulate the clinically relevant loading of a full prosthetic system and observe socket and component structural failure. In some cases, observed material failure modes (tensile, compressive, or shear) can also be qualitatively observed. However, these full-system tests are limited in their ability to quantify materials-level measures of stress and strain without additional measurement techniques being applied.

In addition, each CFAB facility was asked to supply the associated distal components resulting in different distal components between the CFA and the CFB, TPB, and 3DC sockets. As well, the TPB and the 3DC sockets had distal plates that were attached using an epoxy and fiberglass wrap. This method of distal plate attachment introduces a location for potential failure that does not indicate the strength of the actual socket material, but rather the socket system. Failure was hypothesized to occur at this distal plate attachment point. Understanding the US of the socket system provides clinicians with additional information regarding the integrity of the socket. Failure of the epoxy connection indicates that the strength of the thermoplastic socket and 3D-printed socket is greater than that of the connection between the distal plate and the socket. Further testing should explore the behavior of 3D-printed sockets designed for use with a pin-lock system, the hypothesized worst-case design. Replicating the completed testing with this socket design will also provide information for other failure modes for 3D-printed sockets. The differences between the socket fabrication methods limit the direct assessment of how socket material affects overall strength, but it does allow for greater understanding of the behavior of a variety of socket types under load, which is limited in current published work.

CONCLUSIONS

The fit and function of a lower-limb prosthetic socket is crucial to the success of a prosthesis. Although the design of a prosthetic socket has largely been focused on improving comfort for a patient, the strength of the socket should not be overlooked. Sockets are required to withstand severe single instance loads, and socket failure could be detrimental. The ease of customization that accompanies additive manufacturing, specifically 3D-printing, lends well to prosthetic socket design and fabrication. Although the speed and ease of 3D printing are usually not contested, the strength of 3D-printed materials often is, especially for use in prosthetics. Advancements in 3D-printing processes and materials increase the viability for use in prosthetic socket fabrication. However, it is still crucial to analyze the strength of sockets produced using 3D printing. The ISO standard has previously been used to guide strength testing of sockets fabricated using traditional techniques and thus serves as a relevant standard to guide testing of 3D-printed sockets. Using similar test methods allows for comparison between the new 3D-printed sockets and traditional sockets. 3D printing shows great promise as a method to produce custom prosthetic sockets. Continued refinement of socket designs and printing methods will allow for 3D-printed sockets that have adequate strength and improved fabrication methods.

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